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**4th Sem / Mech, CNC, Mech, Adv. Manuf. Tech.,
Mech Engg (CAD / CAM Dsgn & Robotics)
Sub : Thermodynamics-II/I.C. Engines**

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Heat balance sheet of an engine shows:
- a) Fuel price
 - b) Distribution of heat energy
 - c) Engine weight
 - d) Engine size
- Q.2 Which device is used in water cooling systems?
- a) Radiator
 - b) Carburetor
 - c) Silencer
 - d) Flywheel
- Q.3 The cycle used in petrol engines is:
- a) Diesel cycle
 - b) Otto cycle
 - c) Brayton cycle
 - d) Rankine cycle
- Q.4 Which system supplies fuel in a diesel engine?
- a) Carburetor system
 - b) Ignition coil system
 - c) Fuel injection system
 - d) Cooling system

- Q.5 What is the main function of a steam turbine ?
- a) To convert electrical energy into heat energy
 - b) To convert heat energy of steam into mechanical energy
 - c) To convert mechanical energy into heat energy
 - d) To convert steam into water
- Q.6 Brake power of an engine is the power:
- a) At the crankshaft output
 - b) Inside the cylinder
 - c) Lost in exhaust
 - d) Lost in friction
- Q.7 What are the main types of condensers?
- a) Jet condenser and Surface condenser
 - b) Open condenser and Closed condenser
 - c) Axial condenser and Radial condenser
 - d) Rotary condenser and Static condenser
- Q.8 The Ramjet engine works on which principle?
- a) External compressor
 - b) Electric motor
 - c) Steam expansion
 - d) High-speed air compression
- Q.9 Which of the following is not an alternate fuel
- a) CNG
 - b) LPG
 - c) Oxygen
 - d) Hydrogen

Q.10 Which fuel is commonly use in jet propulsion?

- a) Petrol b) Diesel
c) Kerosene d) Coal

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

Q.11 Expand CRDI.

Q.12 Name any one pollutant of IC engine.

Q.13 Cut off ratio is _____.

Q.14 Which Engine has greater compression ratio? (Petrol / diesel)

Q.15 Mechanical efficiency is the ratio of _____.

Q.16 The water jackets are located at_____.

Q.17 Radiator fan gets Power from_____.

Q.18 Name any one impulse turbine.

Q.19 Low volumetric efficiency indicates_____.

Q.20 The secondary coil of ignition coil is made of _____ wire. (Thick / thin)

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 Explain the working of 4-stroke petrol engine.

Q.22 Explain the air-fuel ratios required for various load requirement of engine.

Q.23 Write short note on working of battery ignition system.

- Q.24 Describe the working of fuel feed pump in diesel engine.
- Q.25 Explain the working of reaction turbine with Diagram. Give example.
- Q.26 Enlist the various components in diesel fuel system.
- Q.27 Explain the working of rocket engine.
- Q.28 Explain the working of simple carburetor with its limitations.
- Q.29 Explain the working of thermostat with its diagram.
- Q.30 Explain the function of cooling system in IC engines.
- Q.31 Describe the method of find brake power.
- Q.32 Explain the properties of lubricant.
- Q.33 Compare turbine with reciprocating engine.
- Q.34 Give any five faults in lubrication system with their remedies.
- Q.35 Explain the function and working of:
a) Crank shaft b) Inlet valve

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Explain the difference between Euro-I and Euro-II norms. What are their significance?
- Q.37 Classify steam condensers. Explain the working surface condenser with diagram.
- Q.38 Compare open cycle and closed cycle gas turbine.